Hierarchical vs. Flat CNNs for Object Detection

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Motivation & Description

Biological inspiration: Human visual cortex processes edges → corners → contours → full objects

Curriculum learning: Prior work shows that staged tasks can speed convergence and improve feature quality (Bengio et al. 09)

Gap: Most CNNs learn end-to-end; we ask: *Does explicitly layering low- to high-level representations help recognition?*

Model Architectures & Training

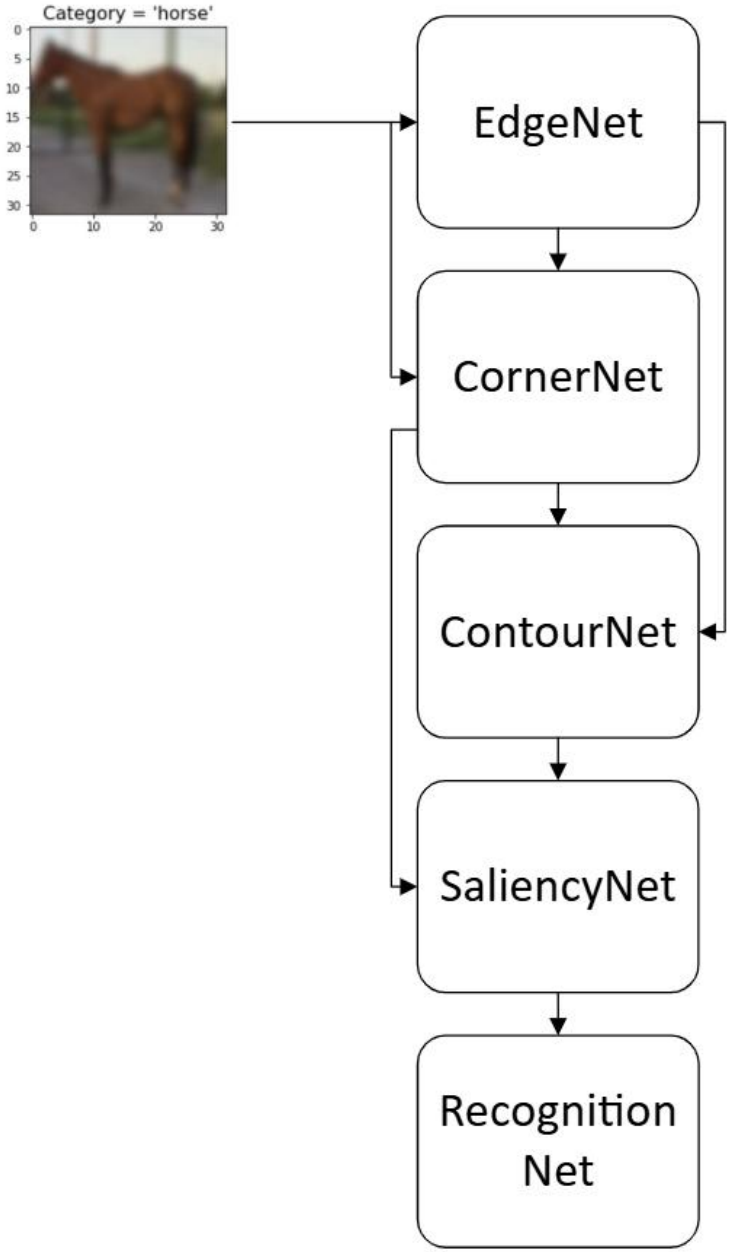
Hierarchical Pipeline (Total ~0.25 M params)

- **EdgeNet** (0.06 M) Accuracy: 96.00 % (pixel)
  - Input: RGB → Sobel-style edge maps
- **CornerNet** (0.05 M) Accuracy in : 87.08 % (pixel)
  - Input: RGB + EdgeMap → corner detections
- **ContourNet** (0.04 M) Accuracy: 65.06 % (pixel)
  - Input: EdgeMap + CornerMap → contour detections
- **SaliencyNet** (0.06 M) Accuracy : 0.000554 (MSE)
  - Input: CornerMap + ContourMap → saliency scores
- **RecognitionNet** (0.04 M)
  - Input: concat(AllMaps) + RGB → 10-way softmax

Flat CNN (0.09 M params)

- Standard 3-layer conv ⇒ pooling ⇒ FC

- **Training Details:**
- **Dataset:** CIFAR-10
- **Optimizer:** Adam, lr=1e-3, batch size=64



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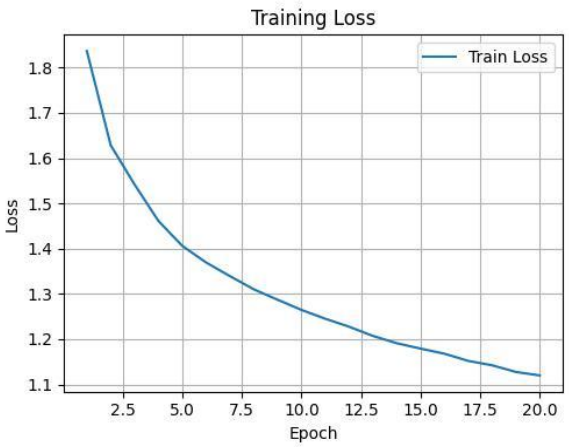
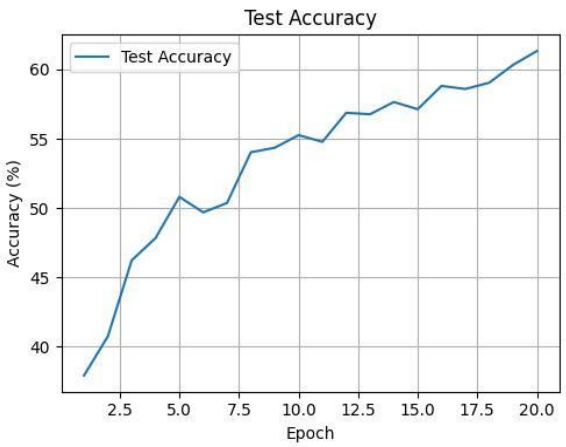
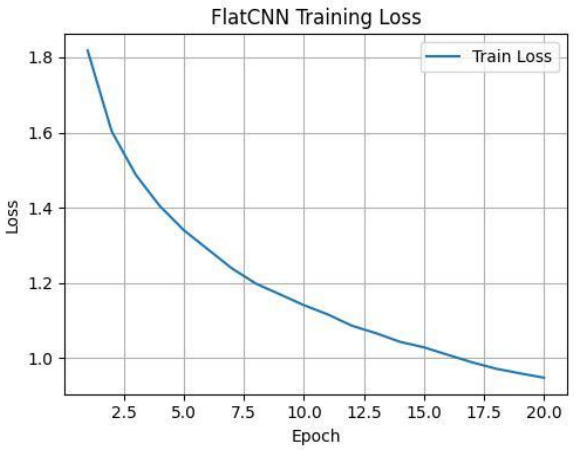
Experimental Results

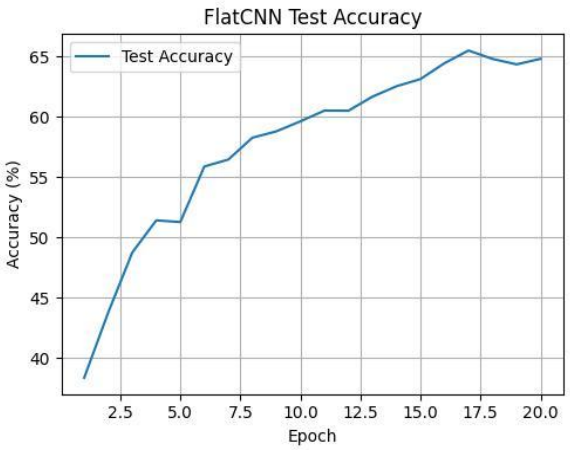
1) Full-Data (CIFAR-10)

Model	Params	10 ep Time	Test Accuracy
Flat CNN	0.09 M	58 s	**63.78 %**
Hierarchical	0.25 M	178 s	61.25 %



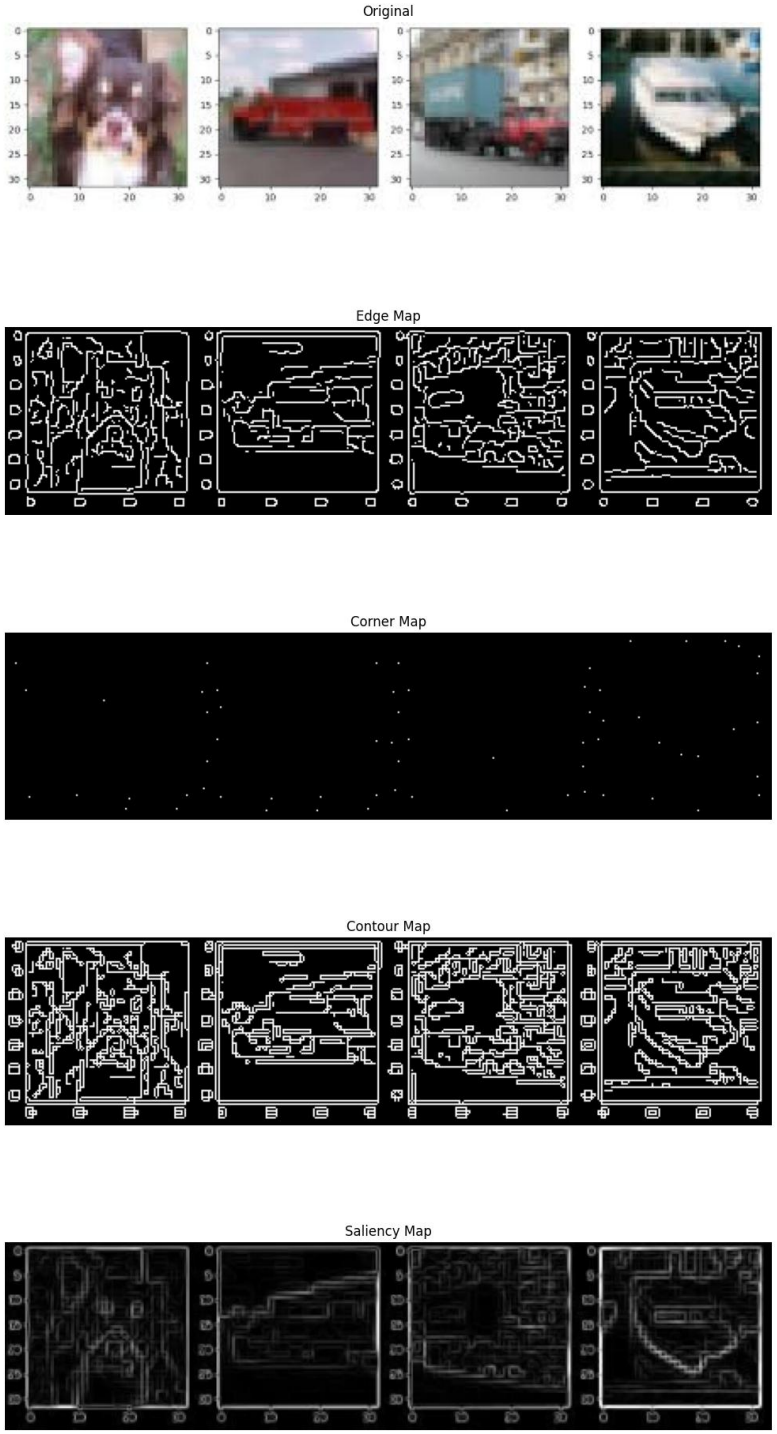






2) Ablation (RGB + Single Map, 20 ep)

- **RGB only:** 57.88 % (3-channel)
- **RGB+Edge:** 57.59 % (4-channel)
- **RGB+Corner:** 59.00 % (4-channel)
- **RGB+Contour:** 56.85 % (4-channel)
- **RGB+Saliency: 59.88 %** (4-channel)
- **RGB+All:** 59.47% (7-channel)



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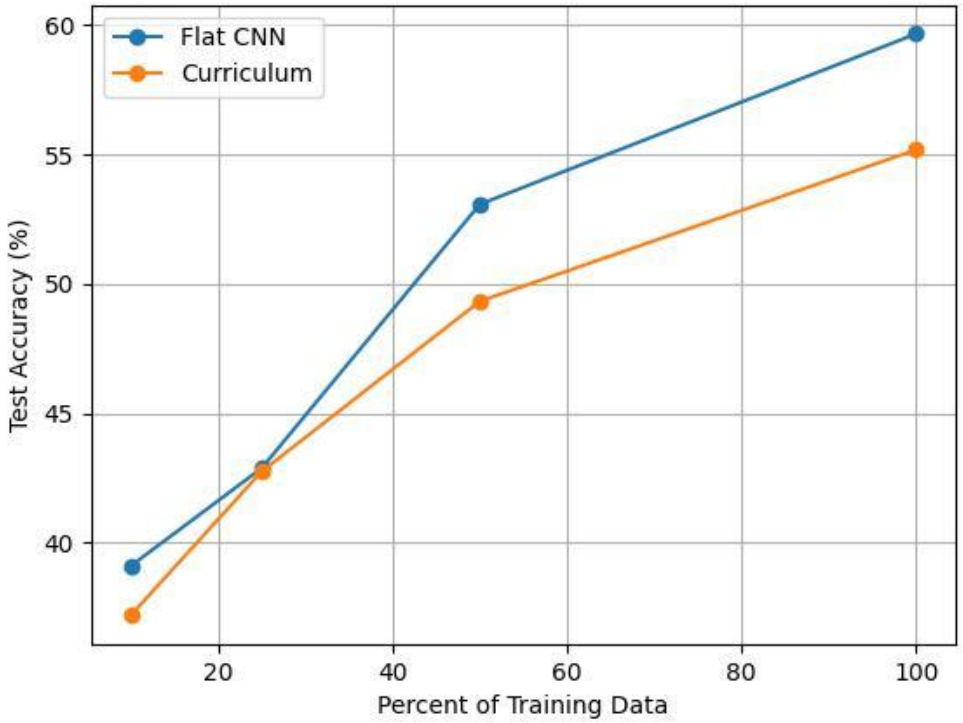
Data-Efficiency Sweep

- At 10 %, 25 %, 50 % of data: FlatCNN leads, but gap shrinks (trained for 10 epochs)



Discussion, Limitations & Future Work

- **Key Takeaways:**
  - **Accuracy:** Flat CNN still best with full data
  - **Interpretability:** Hierarchical maps clearly detect intended features
  - **Efficiency:** Hierarchy narrows gap in low-data regimes

- **Limitations:**
  - More parameters & training time
  - Tested only on clean dataset (CIFAR-10)

- **Future Directions:**
  - **End-to-end fine-tuning:** allow backprop across all modules
  - **Robustness tests:** add noise / domain shift (e.g., TinyImageNet, real-world photos)
  - **Larger datasets:** scale to CIFAR-100 or ImageNet-lite
  - **Further investigations on saliency maps**

References

[1] Yoshua Bengio, Jerome Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In Proceedings of the 26th annual international conference on machine learning, pages 41–48. ACM, 2009.